

Supplementary Material

Top-6-Class MetaStudio (Top6Meta): A Python-based Framework for the Modeling of Architected Materials and Structures

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1. Architected Materials and Structures

1.1. Strut-based Lattice and Structural Designs

The proposed software framework provides a dedicated design module for strut-based lattice architectures, enabling systematic generation, parametric control, and scalable construction of a broad class of strut-lattice topologies for advanced materials and mechanical structures, as elaborated in Figure 1 of the main part of the Manuscript.

Strut-based lattice designs encompass a wide range of structural topologies, including body-centered cubic (BCC), Octet, reinforced Octet (R-Octet), Octahedral, reinforced Octahedral (R-Octahedral), and circular Octahedral (C-Octahedral) architectures, as illustrated in Figure ??a. BCC lattice is composed of struts connecting the center of the cubic unit cell to its eight vertices, resulting in a relatively low coordination number and a predominantly bending-dominated deformation response. The Octet lattice consists of interconnected tetrahedral and octahedral sub-units, forming a highly connected, stretch-dominated architecture characterized by high stiffness and load-bearing efficiency. The reinforced Octet (R-Octet) topology augments the conventional Octet lattice through the introduction of additional vertical struts aligned with the principal loading direction, thereby improving axial stiffness, structural stability, and resistance to buckling. The Octahedral lattice features struts arranged along the edges of octahedral units, leading to intermediate nodal connectivity and mixed bending–stretching deformation modes. Its reinforced variant, the reinforced Octahedral (R-Octahedral) lattice incorporates supplementary vertical member at center that enhance nodal coordination and load transfer continuity under compressive and flexural loading. Finally, the circular Octahedral (C-Octahedral) lattice is derived from the Octahedral

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topology by replacing straight struts with smoothly curved members, which reduces stress concentrations at the nodes, promotes smoother load redistribution, and enhances damage tolerance. Together, these lattice architectures span a wide range of connectivity and deformation mechanisms, providing a versatile design space for topology-driven optimization within the proposed software framework.

For each topology, the unit-cell geometry is described using a unified set of parametric variables, namely the unit cell length (l_u), strut diameter (d), strut length ($2l$), and topology-dependent orientation angles (θ). The local relative density (or equivalently, volume fraction) ρ^* of a strut-based lattice is expressed as a function of the strut slenderness ratio, i.e., $\rho^* = f(d/l_u)$, where the specific functional form depends on the unit-cell connectivity and strut orientation. For the body-centered cubic (BCC) topology, the geometric parameters are explicitly related through closed-form trigonometric relations. In particular, the lattice angle is fixed by the BCC geometry as $\theta = \sin^{-1}(1/3)$, while the strut length is related to the unit-cell size by $2l = \sqrt{3} l_u$. These relations uniquely define the strut orientation and length for a given unit-cell size, thereby ensuring periodic compatibility and exact nodal connectivity upon tessellation.

For a given topology, analytical relations between l_u , l , and θ are enforced to preserve periodicity and exact node matching across adjacent unit cells. While the specific strut orientations and connectivity differ across lattice families, the common parametric formulation enables consistent control over the relative density, directional stiffness, and load-bearing pathways. This generalized design approach facilitates systematic comparisons among different lattice topologies and allows their seamless integration within the proposed software framework for scalable design and additive manufacturing of architected structures. Furthermore, the framework supports unit-cell-based geometric gradation by allowing controlled variation of the governing parameters along any prescribed Cartesian direction, thereby enabling the design of functionally graded strut-type lattice architectures with spatially tailored mechanical responses.

1.2. Triply Periodic Minimal Surface (TPMS)–Based Architected Materials and Structures

The present software framework also provides dedicated design modules for skeletal and sheet type triply periodic minimal surface (TPMS) architectures, enabling implicit representation, parametric control, scalable generation of complex architected geometries, and seamless embedding within cubic and cylindrical domains as well as beam-, plate-, sandwich-, and column-like structural configurations.

Triply periodic minimal surface (TPMS) architectures are generated using implicit level-set representations of the form $\phi_{TPMS}(x, y, z) = t$, where the scalar field ϕ_{TPMS} defines the surface morphology and the threshold t controls the resulting volume fraction. The framework supports a broad range of TPMS and TPMS-related geometries, including Gyroid, Diamond, Primitive, Neovius, IWP, Fischer–Koch, Lidinoid, and related variants, whose analytical level-set equations are summarized in Table 1. By appropriate selection of the level-set formulation and thresholding strategy, both sheet-type and skeletal TPMS architectures can be constructed for a desired volume fraction and number of unit cell repetitions along the x , y and z directions.

1.3. Spinodal Decomposition–Based Stochastic Architected Designs

Spinodal decomposition–based stochastic architectures are incorporated into the framework as a distinct design class that enables the generation of irregular, non-periodic architected materials with controllable morphological characteristics. Unlike strut-based and TPMS-based designs, these architectures are governed by statistically defined fields, allowing complex, spatially continuous microstructures to be generated without explicit unit-cell periodicity. The framework provides parametric control over key stochastic descriptors, including characteristic length scale, phase connectivity, and morphological anisotropy, allowing systematic tuning of the resulting architecture. The stochastic module supports a range of dimensional configurations, including one- and two-dimensional spinodal patterns as well as fully three-dimensional skeletal and sheet-like morphologies. These design options enable the generation of architectures spanning layered and columnar stochastic structures to fully interconnected three-dimensional networks, providing broad flexibility for tailoring mechanical and functional performance. Representative stochastic architectures generated using this module are illustrated in Figure 1 of the main part of the manuscript. Stochastic topologies originate from phase separation phenomena governed by the Cahn–Hilliard equation: [1, 2]:

$$\frac{\partial c}{\partial t} = D \nabla^2 (c^3 - \omega \nabla^2 c - c), \quad (1)$$

where c represents the concentration of the two phases, D is the diffusion coefficient, and ω controls the thickness of the interfacial region between phases. Although physically representative

Table 1: Level-set equations of selected TPMS and related surfaces.

Design	Level-set Equation $\varphi_{TPMS}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = t$
Gyroid	$\varphi_{GY}(x, y, z) : \sin x \cos y + \sin y \cos z + \sin z \cos x = t$
Diamond	$\varphi_{DIAM}(x, y, z) : \cos x \cos y \cos z - \sin x \sin y \sin z = t$
Primitive	$\varphi_{PCell}(x, y, z) : \cos x + \cos y + \cos z = t$
Neovius	$\varphi_{NEO}(x, y, z) : 3(\cos x + \cos y + \cos z) + 4 \cos x \cos y \cos z = t$
IWP	$\varphi_{IWP}(x, y, z) : 2(\cos x \cos y + \cos y \cos z + \cos z \cos x) - (\cos 2x + \cos 2y + \cos 2z) = t$
Fischer–Koch	$\varphi_{FKS}(x, y, z) : (\cos x \cos y + \cos y \cos z + \cos z \cos x) - (\cos 2x + \cos 2y + \cos 2z) = t$
I2–Y	$\varphi_{I2Y}(x, y, z) : -2 \sin 2x \cos y \sin z + \sin x \sin 2y \cos z + \cos x \sin y \sin 2z$ $+ \cos 2x \cos 2y + \cos 2y \cos 2z + \cos 2x \cos 2z = t$
FRD Prime	$\varphi_{FRD}(x, y, z) : 4 \cos x \cos y \cos z - (\cos 2x \cos 2y + \cos 2y \cos 2z + \cos 2z \cos 2x) = t$
Lidinoid	$\varphi_{LDN}(x, y, z) : \sin 2x \sin z \cos y + \sin 2y \sin x \cos z + \sin 2z \sin y \cos x - \cos 2x \cos 2y$ $+ \cos 2y \cos 2z + \cos 2z \cos 2x + 0.3 = t$
Split P	$\varphi_{SPP}(x, y, z) : 1.1 \sin 2x \sin z \cos y + \sin 2y \sin x \cos z + \sin 2z \sin y \cos x$ $- 0.2 \cos 2x \cos 2y + \cos 2y \cos 2z + \cos 2z \cos 2x$ $- 0.4 \cos x + \cos y + \cos z = t$

(Eq. (1)), this formulation is computationally demanding and offers limited direct control over the geometric characteristics of the spinodal decomposition. An alternative and more flexible approach approximates the phase field $\phi_{\text{spin}}(\mathbf{X})$, with a Gaussian random field of the following form [3, 4]:

$$\phi_{\text{spin}}(\mathbf{X}) = \sqrt{\frac{2}{N}} \sum_{i=1}^N \cos(\boldsymbol{\kappa}_i \cdot \mathbf{X} + \mu_i), \quad (2)$$

where N is the number of superimposed waves, $\boldsymbol{\kappa}_i$ are wave vectors of equal amplitude and $\mu_i \in \mathbf{U}[0, 2\pi]$ are uniformly distributed random phase shifts. When the wave vectors are sampled uniformly from the unit sphere, the resulting spinodal architectures $\phi_{\text{spin}}(\mathbf{X})$ in Eq. (2) become statistically homogeneous, exhibiting consistent statistical properties throughout the spatial domain. The wave vectors $\boldsymbol{\kappa}_i$ are randomly sampled from the unit sphere ($\mathbf{n}_i \in \mathbb{S}^2$), while the resulting phase field $\phi_{\text{spin}}(\mathbf{X})$ can be thresholded to yield either sheet-like or skeletal-type stochastic networks [5].

Importantly, both TPMS and spinodal architectures are treated within a unified implicit-field framework, facilitating their integration with functional grading, hybrid multimorphology designs, and interpenetrating phase composite generation, enabling their seamless integration into complex architected structures for advanced structural and multifunctional applications.

1.4. Functionally Graded Strut, TPMS, and Stochastic Architectures

The framework provides comprehensive support for the generation of functionally graded architected materials across strut-based, TPMS-based, and stochastic lattice classes. Functional grading is implemented through a unified parametric formulation that allows spatial variation of key geometric descriptors while preserving global shape integrity and topological continuity. Both non-graded (uniform) and graded configurations are supported, allowing direct comparison between constant-density and spatially varying architectures within a consistent design environment. Within the framework, functional gradation (FG) is defined through the spatial variation of the local relative density (equivalently, volume fraction) ρ^* as a function of position. The graded relative density is constrained to the interval $\rho^* \in [\rho_{\min}^*, \rho_{\max}^*]$ and varies along a prescribed grading direction through a normalized spatial coordinate $\xi \in [0, 1]$. A *linear grading law* (FG Type 1) produces a monotonic variation of $\rho^*(\xi)$ along the grading direction, which may be expressed as follows:

$$\rho^*(\xi) = \rho_{\max}^* - (\rho_{\max}^* - \rho_{\min}^*) \xi, \quad (3)$$

and is well suited for components subjected to gradually varying stress fields. This capability enables systematic exploration of functionally graded architected materials. Representative graded architectures and grading profiles are schematically illustrated in Figure 1 of the main part of the Manuscript.

At the macroscale, the external geometry of the architected domain can follow either cubic or cylindrical shapes, thereby accommodating beam-, plate-, and column-like components commonly encountered in structural applications. Functional grading can be selectively activated or deactivated through a dedicated grading flag, allowing a single architectural definition to be used in either uniform or graded form. When grading is enabled, spatial variation may be applied along

any principal Cartesian direction (X, Y, or Z), ensuring compatibility with directional loading conditions and manufacturing constraints.

1.5. Multi-morphology, Hybrid Materials and Structures

In addition to functionally graded designs, the elaborated software framework supports the systematic generation of multi-morphology hybrid architected materials, in which distinct skeletal or sheet-based topologies can be smoothly blended within a single structural design. This approach allows different morphological classes to coexist spatially, thereby enabling localized tailoring of the mechanical response, multifunctionality, and failure behavior beyond what can be achieved through classical or graded designs. Hybrid architected materials are constructed by smoothly transitioning between two distinct implicit topologies through spatial blending of their corresponding level-set fields. In the present framework, linear multimorphology hybridization is achieved using the implicit formulation:

$$\phi_{\text{Hybrid}}(x, y, z) = \alpha(x, y, z) \phi_{TPMS_1}(x, y, z) + (1 - \alpha(x, y, z)) \phi_{TPMS_2}(x, y, z), \quad (4)$$

where ϕ_{TPMS_1} and ϕ_{TPMS_2} denote the level-set functions of the two parent architectures. The transition between the two morphologies is governed by the blending function $\alpha(x, y, z)$ defined as follows:

$$\alpha(x, y, z) = \frac{1}{1 + e^{kG(x, y, z)}}. \quad (5)$$

where the parameter k controls the transition width (large k gives a sharp interface; small k gives a smooth, wide interface), and $G(x, y, z)$ prescribes the spatial position and shape of the transition region. Sharper transitions (larger k , e.g. $k = 10$) tend to induce higher stress concentrations and localized plastic strain near the interface; smoother transitions (e.g. $k = 1$ or $k = 0.5$) mitigate these effects while preserving characteristic topological features. The principal hybridization scheme employs the following transition function G :

$$\text{Linear (1D) along } z: \quad G(x, y, z) = z - \frac{L}{2} \quad (6)$$

Where in Eq. (6), L sets the reference position of the transition (by default it is $L/2$). Substituting the above G functions into (4)–(5) yields the explicit hybrid equations used in this work:

Linear (1D):

$$\phi_{\text{Linear (1D)}}(x, y, z) = \left(\frac{1}{1 + e^{k(z-L/2)}} \right) \phi_{TPMS^1} + \left(1 - \frac{1}{1 + e^{k(z-L/2)}} \right) \phi_{TPMS^2} \quad (7)$$

1.6. Interpenetrating Phase Composites

In addition to classical strut-, stochastic, and TPMS-based architectures, as well as their functionally graded and hybrid multi-morphology designs, the elaborated software framework provides dedicated capabilities for the generation of *interpenetrating phase composite* (IPC) architectures within each of the aforementioned design modules, as schematically illustrated in Figure 1(d) and Figure 1(f) of the main part of the manuscript. Within each design module, —strut-based, TPMS-based, stochastic, and hybrid— the extraction of *both complementary phases* from a common

implicit representation is feasible, resulting in fully co-continuous solid–solid or solid–void architectures. This approach allows architected reinforcement phases to be naturally embedded within a complementary soft or secondary phase, forming mechanically interlocked IPC structures.

IPC generation is performed through analytical Boolean operations applied directly to the underlying implicit fields, ensuring exact phase complementarity and geometric consistency at the interface. Each phase can be independently exported as a high-quality STL file, facilitating seamless integration into *multi-material additive manufacturing* workflows, as well as in *finite element modeling* environments for numerical analysis. This capability enables the reinforcement and matrix phases to be treated as distinct entities, while fully preserving their interpenetrating phase topology.

Importantly, the framework supports seamless integration of IPC generation with *functional grading* and *hybridization*, enabling spatially varying IPC architectures and multi-topology interpenetrating designs within a single modeling pipeline. To the authors’ best knowledge, this represents one of the first software frameworks to provide an integrated and automated capability for generating *multi-STL IPC geometries* suitable for both *multi-material 3D printing* and *numerical simulation*, without the need for manual post-processing or external Boolean operations. As such, the framework significantly expands the accessible design space for architected interpenetrating composites and facilitates their systematic exploration for advanced structural and multifunctional applications. In addition to hybrid multimorphology designs, the framework supports generation of interpenetrating phase composite architectures. By extracting both complementary phases of a given implicit field, co-continuous solid–solid or solid–void architectures can be generated, enabling the construction of IPCs reinforced by TPMS, spinodal, or hybrid cores. This unified implicit formulation allows hybridization, grading, and IPC generation to be combined seamlessly within a single modeling workflow.

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